Course Document

Course Code	UE24CS2401
Course Title	**Artificial intelligence & Machine Learning (AIML)**
Program Code	CS
Program Title	B. Tech. Computer Science and Engineering
School Code	01
School Title	School of Computer Science and Technology
Department Code	CSE
Department	Department of Computer Science and Engineering
Faculty Code	E
Faculty Title	Faculty of Engineering and Technology
Department offering the Course	Computer Science and Engineering
Faculty Member	
Semester Duration	Weeks (1-16) -Teaching, Learning and Continuous Assessment Weeks (17-18) -SEE Weeks (19-20)- Announcement of Results

1. Course Size

Credits	L	T	P	Hours/Week
3	**2**	**0**	**2**	**3**

Total Term/ Semester hours: 45

2. Course Details

2.1 Course Aims and Summary

- This course introduces students to the foundations of Artificial Intelligence (AI) and Machine Learning (ML). It begins with basic AI concepts such as intelligent agents, search algorithms, and knowledge representation. The course then covers reasoning techniques and probabilistic models.
- Students are introduced to machine learning principles, data understanding, feature engineering, and dimensionality reduction. Core learning models including regression, decision trees, Bayesian learning, clustering, and neural networks are discussed.
- The course emphasizes conceptual clarity, problem-solving ability, and practical understanding of AI and ML techniques used in real-world applications.

2.2 Course Objectives

The objectives of the Course are:

- Understand the fundamentals of Artificial Intelligence, including intelligent agents, environments, and informed search algorithms.
- Explain knowledge representation techniques and reasoning mechanisms used in AI systems.
- Understand the need and scope of Machine Learning and its relationship with other computing disciplines.
- Analyze and preprocess data using basic data analysis, visualization, and feature engineering techniques.
- Apply similarity-based and regression learning methods to solve prediction problems.
- Understand decision tree learning models and their construction using ID3 algorithm.
- Apply probabilistic learning techniques, including Bayesian learning and Naïve Bayes classification.
- Understand unsupervised learning methods, particularly clustering using the K-means algorithm.
- Explain the fundamentals of artificial neural networks, including perceptron learning and basic neural models.
- Develop the ability to select appropriate AI/ML techniques for simple real-world problem scenarios.

2.3 Course Outcomes

After undergoing this course students will be able to:

CO1	*Define* the basic concepts, terminology, and principles of Artificial Intelligence and Machine Learning, including intelligent agents, search algorithms, knowledge representation, reasoning, data understanding, and fundamental learning models such as supervised and unsupervised.
CO2	*Describe* the fundamental concepts of Artificial Intelligence and Machine Learning, including intelligent agents, search algorithms, knowledge representation, reasoning, data understanding, and basic learning models.
CO3	*Apply* Artificial Intelligence and Machine Learning techniques, including search algorithms, knowledge representation, and basic learning models, to solve simple real-world problems.

CO4	*Analyze* problems and datasets to select appropriate Artificial Intelligence and Machine Learning techniques, including search methods, knowledge representation, and basic learning models.
CO5	*Evaluate* Artificial Intelligence and Machine Learning techniques to determine their suitability and effectiveness
CO6	*Design* a simple Artificial Intelligence or Machine Learning–based solution by integrating appropriate search, reasoning, and learning techniques.

Outcome Map:

COs	PO 01	PO 02	PO 03	PO 04	PO 05	PO 06	PO 07	PO 08	PO 09	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1	**3**							**3**	**3**			**2**			
CO2	**2**	**3**						**3**	**3**			**2**			
CO3	**2**	**2**	**2**		**2**			**2**	**3**			**2**	**2**	**2**	
CO4	**2**	**2**	**2**		**2**			**2**	**3**			**2**	**2**	**1**	
CO5	**2**	**1**	**1**	**2**	**1**			**2**	**2**	**2**	**2**	**2**	**1**	**2**	**1**
CO6	**1**	**1**	**1**	**2**	**1**	**3**		**2**	**2**	**2**	**2**	**2**	**1**	**1**	**1**

Relevance: 1 high, 2 medium, 3 low

2.4 Course Content:

1. **Introduction to Artificial intelligence:** What is AI, History of AI, Types of AI, Components of Intelligence, Agent and Environment, Informed Search Algorithms: Pure Heuristic, Best first, A*, AO* Algorithms.
2. **Knowledge Representation:** Introduction, Knowledge representation, Knowledge based agent, types of knowledge, Techniques of knowledge representation in AI, Syntax.

   Reasoning and Learning in AI: Reasoning, Probabilistic Reasoning in AI.
3. **Introduction to Machine Learning**
   - **Introduction:** Need for Machine Learning, Machine Learning Explained, Machine Learning in Relation to other Fields, Types of Machine Learning.
   - **Understanding Data**: Introduction, Univariate Data Analysis and Visualization. Bivariate Data and Multivariate Data, Feature Engineering and Dimensionality Reduction Techniques.
4. **Basic Learning Theory**: Design of Learning System, Introduction to Concept of Learning, Modelling in Machine Learning.

   Similarity-based Learning: Nearest-Neighbor Learning, Weighted K-Nearest-Neighbor Algorithm.

   Regression Analysis: Introduction to Regression, Introduction to Linear Regression, Multiple Linear Regression.

   Decision Tree Learning: Introduction to Decision Tree Learning Model, Decision Tree Induction Algorithms: ID3 Tree Construction.
5. **Bayesian Learning**: Introduction to Probability-based Learning, Fundamentals of Bayes Theorem and Classification Using Bayes Model: Naïve Bayes Algorithm.

   Clustering Algorithms: Introduction to Clustering Approaches, K-means Algorithm

   Artificial Neural Networks: Introduction, Biological Neurons, Artificial Neurons, Perceptron and Learning Theory: XOR, Types of Artificial Neural Networks.

Laboratory Programs:

1. Design and implement the A* search algorithm for a given weighted graph. The algorithm should compute the shortest path from a start node to a goal node using:

   g(n): Cost from start node to current node

   h(n): Heuristic estimate from current node to goal

   f(n) = g(n) + h(n).

2. Develop a program to create histograms for all numerical features and analyze the distribution of each feature. Use California Housing dataset.

3. Develop a program to implement Principal Component Analysis (PCA) for reducing the dimensionality of the Iris dataset from 4 features to 2.

4. For a given set of training data examples stored in a .CSV file, implement and demonstrate the Find-S algorithm to output a description of the set of all hypotheses consistent with the training examples.

5. Develop a program to implement k-Nearest Neighbour algorithm to classify the randomly generated 100 values of x in the range of [0,1]. Perform the following based on dataset generated.
   a) Label the first 50 points {x1,……,x50} as follows: if ($xi \leq 0.5$), then $xi \in$ Class1, else $xi \in$ Class1
   b) Classify the remaining points, x51,……,x100 using KNN. Perform this for k=1,2,3,4,5,20,30

6. Develop a program to demonstrate the working of Linear Regression. Use Boston Housing Dataset for Linear Regression.

7. Develop a program to demonstrate the working of the decision tree algorithm. Use Breast Cancer Data set for building the decision tree and apply this knowledge to classify a new sample.

8. Develop a program to implement k-means clustering using Wisconsin Breast Cancer data set and visualize the clustering result.

2.5 Course Recourses

Textbooks:

- Reema Thereja, “Artificial Intelligence –Beyond Classical AI”, Pearson,2023.
- S Sridhar and M Vijayalakshmi, “Machine Learning” , Oxford University Press ,2021.

3.Teaching and Assessment

3.1 Teaching

Lecture Number	Lecture Topic	Lecture Slides	Lecture Videos
0	Introduction to Artificial intelligence and Introduction to Machine Learning	Lecture-00	Video-00
1	Introduction to Artificial intelligence: What is AI, History of AI, Types of AI	Lecture-01	Video-01
2	Components of Intelligence, Agent and Environment	Lecture-02	Video-02
3	Components of Intelligence, Agent and Environment	Lecture-03	Video-03
4	Informed Search Algorithms: Pure Heuristic	Lecture-04	Video-04
5	Best first, A*	Lecture-05	Video-05
6	Best first, A* + Lab Program 1	Lecture-06	Video-06
7	AO* Algorithms	Lecture-07	Video-07
8	AO* Algorithms	Lecture-08	Video-08
9	Knowledge Representation: Introduction, Knowledge representation	Lecture-09	Video-09
10	Knowledge Representation: Introduction, Knowledge representation,	Lecture-10	Video-10
11	Knowledge based agent, types of knowledge,	Lecture-11	Video-11
12	Techniques of knowledge representation in AI, Syntax	Lecture-12	Video-12
13	Techniques of knowledge representation in AI, Syntax	Lecture-13	Video-13
14	Reasoning and Learning in AI: Reasoning, Probabilistic Reasoning in AI.	Lecture-14	Video-14
15	Lab Program 2	Lecture-15	Video-15
16	Reasoning and Learning in AI: Reasoning, Probabilistic Reasoning in AI.	Lecture-16	Video-16
Quiz -01 and Test-1-Obtain Student Feedback			
17	Introduction to Machine Learning • Introduction: Need for Machine Learning, Machine Learning Explained	Lecture-17	Video-17
18	Machine Learning in Relation to other Fields, Types of Machine Learning.	Lecture-18	Video-18
19	Understanding Data: Introduction, Univariate Data Analysis and Visualization.	Lecture-19	Video-19

20	Lab Program 3	Lecture-20	Video-20
21	Bivariate Data and Multivariate Data,Feature Engineering and Dimensionality Reduction Techniques	Lecture-21	Video-21
22	Feature Engineering and Dimensionality Reduction Techniques	Lecture-22	Video-22
23	Basic Learning Theory: Design of Learning System, Introduction to Concept of Learning, Modelling in Machine Learning	Lecture-23	Video-23
24	Similarity-based Learning: Nearest-Neighbor Learning, Weighted K-Nearest-Neighbor Algorithm	Lecture-24	Video-24
Submission of Assignment-1			
25	Similarity-based Learning: Nearest-Neighbor Learning, Weighted K-Nearest-Neighbor Algorithm + Lab program 4	Lecture-25	Video-25
26	Regression Analysis: Introduction to Regression, Introduction to Linear Regression, Multiple Linear Regression.	Lecture-26	Video-26

27	Regression Analysis: Introduction to Regression, Introduction to Linear Regression, Multiple Linear Regression.	Lecture-27	Video-27
28	Decision Tree Learning: Introduction to Decision Tree Learning Model, Decision Tree Induction Algorithms: ID3 Tree Construction.	Lecture-28	Video-28
29	Decision Tree Learning: Introduction to Decision Tree Learning Model, Decision Tree Induction Algorithms: ID3 Tree Construction. + Lab Program 5	Lecture-29	Video-29
30	Bayesian Learning: Introduction to Probability-based Learning, Fundamentals of Bayes Theorem	Lecture-30	Video-30
31	Bayesian Learning: Introduction to Probability-based Learning, Fundamentals of Bayes Theorem	Lecture-31	Video-31
32	Lab Program 6	Lecture-32	Video-32
33	Classification Using Bayes Model: Naïve Bayes Algorithm	Lecture-33	Video-33
Quiz -02 and Test -02			
34	Classification Using Bayes Model: Naïve Bayes Algorithm	Lecture-34	Video-34
35	Classification Using Bayes Model: Naïve Bayes Algorithm	Lecture-35	Video-35
36	Clustering Algorithms: Introduction to Clustering Approaches, K-means Algorithm	Lecture-36	Video-36

37	Lab Program 7	Lecture-37	Video-37
38	Clustering Algorithms: Introduction to Clustering Approaches, K-means Algorithm	Lecture-38	Video-38
39	Clustering Algorithms: Introduction to Clustering Approaches, K-means Algorithm	Lecture-39	Video-39
40	Artificial Neural Networks: Introduction, Biological Neurons, Artificial Neurons	Lecture-40	Video-40
41	Artificial Neural Networks: Introduction, Biological Neurons, Artificial Neurons	Lecture-41	Video-41
42	Perceptron and Learning Theory: XOR,	Lecture-42	Video-42
43	Types of Artificial Neural Networks.	Lecture-43	Video-43
44	Types of Artificial Neural Networks	Lecture-44	Video-44
45	Lab Program 8	Lecture-45	Video-45
	Quiz-03 and Test-03 Submission of Assignment 2 **Obtain Student Feedback**		
	Examination Preparation Break		
	Term/Semester End Examination		

3.2. Assessment weight Distribution

	Quiz	Test	Assignment/ PBL/PrBL	SEE	Total Marks
Weights/ Course Outcomes	15	25	20	40	100
CO1	10			4	14
CO2	5	5		10	20
CO3		9		11	20
CO4		9		8	17
CO5		2	10	3	15
CO6			10	4	14

3.3. Schedule of Assessment

Assessment Type	Dates	Marks	COs	Quiz	Test	Assignment/ PBL/ PrBL	SEE
Weight				15	25	20	40
Duration				30 min	60 min	6 weeks	3 hours
Quiz-1	5th week	6	CO1/CO2				
Quiz-2	10th week	5	CO3/ CO4				
Quiz-3	15th week	4	CO5/ CO6				
Test-1	5th week	10	CO1/CO2				
Test-2	10th week	8	CO3/ CO4				
Test-3	15th week	7	CO5/ CO6				
Assignment-1	7th week	09	CO 1-3				
Assignment-2	14th week	11	CO 4-6				
SEE	18th Week	40	All				

3.4. Grading Criterion

- Based on total marks scored grade is Awarded.

  If marks scored is:

  - 91 and above O (outstanding); 81-90 : A+ (Excellent); 71-80: A (Very Good); 61-70: B+ (Good); 51-60 : B (Above Average); 40 -50: C (Average); below 40: D (Not satisfactory)
- If one scores D grade, the candidate is required to re-register for the course if he/she wants to earn the credit at his/her own convenience

Attainment Calculations:

Recording Marks and Awarding Grades

S. No.	USN	Student Name	Quiz (15%)	Test (25%)	Assignment 20%	SEE 40%	Marks Scored	Grade obtained
1								
2								
3								
N								
Total							XXXXX	

Class Average Marks: Total marks of All Students (XXXX)/ Number of students (N)

Average Grade:

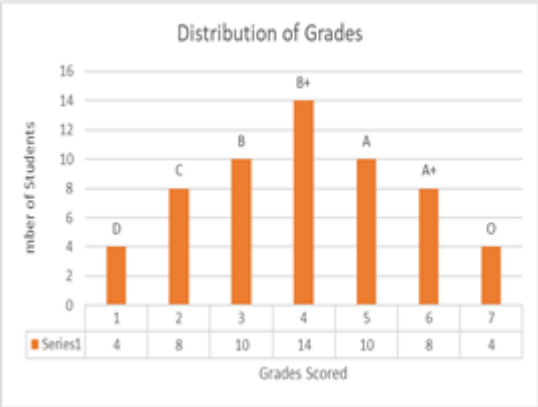
Setting Attainment Targets:

Attainment of Course Outcomes-COs	
Outcomes- Targeted	**Targeted Attainment Level**
70% of students will score C grade and above - Attainment Level 1 60% of students will score C grade and above - Attainment Level 2 50% of students will score C grade and above - Attainment Level 3	1
70% of students will score C grade and above - Attainment Level 1 60% of students will score C grade and above - Attainment Level 2 50% of students will score C grade and above - Attainment Level 3	1
70% of students will score C grade and above - Attainment Level 1 60% of students will score C grade and above - Attainment Level 2 50% of students will score C grade and above - Attainment Level 3	1
70% of students will score C grade and above - Attainment Level 1 60% of students will score C grade and above - Attainment Level 2 50% of students will score C grade and above - Attainment Level 3	1
70% of students will score C grade and above - Attainment Level 1 60% of students will score C grade and above - Attainment Level 2 50% of students will score C grade and above - Attainment Level 3	1
70% of students will score C grade and above - Attainment Level 1 60% of students will score C grade and above - Attainment Level 2 50% of students will score C grade and above - Attainment Level 3	1

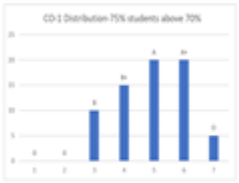
Performance Recording

Academic Year 2023-24	Program: B.Tech., in Computer Science and Engineering	Semester I	**Section** A	**Course Code** XXYYZZH11	**Course Title** Programming with C **Course Tutor/s:** **Tutor's ID/Department:**					
Total Number of students in the Class	Number of Students appeared for all the components of Assessment	Number of Students - Passed all the component of Examination	Class Average Marks	O-Graders >= 91	A+ Graders 81<=M<=90	A Grader 71<=M<=80	B+ Graders 61<=M<=70	B Graders 51<=M<=60	C Graders 40<=M<=50	D Graders M<40
60	58	54	58 B Grade	4	8	10	14	10	8	4
CO1- Performance										
CO2- Performance										
CO3- Performance										
CO4- Performance										
CO5- Performance										
CO6- Performance										

Performance Plotting



CO-1 Distribution



CO-2 Distribution

CO-3 Distribution

CO-4 Distribution

CO-5 Distribution

CO-6 Distribution

Mapping of Course Outcomes with Program Outcome

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1															
CO2															
CO3															
CO4															
CO5															
CO6															

4. Other Details

4.1 Assignment Details or Problem Based Learning: Assignments will be given at the beginning of each block period and students can continuously work on assignment and submit at the end of the block period as per the format provided.

4.2. Academic Integrity Policy: Students are required to strictly follow academic honesty and Integrity: Copying and plagiarism in any form for any of the assessment components will result in Zero .